

Timothy C. Peter
Site Vice President- JAF

JAFP-23-0006
January 31, 2023

United States Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, D.C. 20555-0001

James A. FitzPatrick Nuclear Power Plant
Renewed Facility Operating License No. DPR-059
NRC Docket No. 50-333

Subject: LER: 2022-002-01, Mode Switch Failed to Bypass Low Main Steam Line
in Mode 2 Resulting in MSIV and RPS System Actuation

Reference: FitzPatrick letter, LER: 2022-002, Mode Switch Failed to Bypass Low
Main Steam Line in Mode 2 Resulting in MSIV and RPS System
Actuation, JAFP-22-0048, dated November 22, 2022

Dear Sir or Madam:

This report is being submitted pursuant to 10 CFR 50.73(a)(2)(iv)(A).

There are no new regulatory commitments contained in this report.

Questions concerning this report may be addressed to Mr. Mark Hawes, Regulatory Assurance
Manager (Acting), at (315) 349-6659.

Sincerely,



Timothy C. Peter
Site Vice President

TCP/mh

Enclosure: LER: 2022-002-01, Mode Switch Failed to Bypass Low Main Steam Line
in Mode 2 Resulting in MSIV and RPS System Actuation

cc: USNRC, Region I Administrator
USNRC, Project Manager
USNRC, Resident Inspector
INPO Records Center (IRIS)

NRC FORM 366 (08-2020)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023 Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.					
 LICENSEE EVENT REPORT (LER) (See Page 3 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)										
1. Facility Name James A. FitzPatrick Nuclear Power Plant					2. Docket Number 05000333		3. Page 1 OF 4			
4. Title Mode Switch Failed to Bypass Low Main Steam Line in Mode 2 Resulting in MSIV and RPS System Actuation										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	26	2022	2022 -	002 -	01	01	31	2023	N/A	N/A
9. Operating Mode 2						10. Power Level 000				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(4)		
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.71(a)(5)		
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(1)		
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(i)		
☐ 20.2203(a)(2)(iii)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.77(a)(2)(ii)		
☐ 20.2203(a)(2)(iv)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)				
☐ 20.2203(a)(2)(v)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)				
☐ OTHER (Specify in Abstract below or in NRC Form 366A).										
12. Licensee Contact for this LER										
Licensee Contact Mr. Mark Hawes, Regulatory Assurance Manager (Acting)								Telephone Number (Include Area Code) 315-349-6659		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS	
D	JC	HS	G080	Yes						
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☒ No		☐ Yes (If yes, complete 15. Expected Submission date)			_____			_____	_____	_____
Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)										
On September 26, 2022, at 0306 EDT, while the reactor was sub-critical and inserting control rods for a planned refueling outage, James A. FitzPatrick Nuclear Power Plant (JAF) experienced a reactor scram and Group 1 isolation on low steam line pressure logic. As a result of Group 1 isolation signal seven out of eight Main Steam Isolation Valves (MSIV) isolated. 'A' outboard MSIV (29AOV-86A) failed to isolate on the initial signal. The Reactor Protection System (RPS) actuated, resulting in all control rods being fully inserted, and manual actuation of RCIC. This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A). The RPS actuation was determined to be a failure of the reactor mode switch (05A-S1) to enact the logic required to bypass MSIV isolation on low main steam line pressure and RPS actuation on MSIV closed position. Operating Procedure OP-65 was revised to ensure proper switch contact logic is validated by redundant indications.										



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
James A. FitzPatrick Nuclear Power Plant	05000 – 333	YEAR 2022	SEQUENTIAL NUMBER – 002	REV NO. – 01

NARRATIVE

Background

The Reactor Mode Switch (05A-S1) controls the operating Plant Mode of the Reactor. It has four positions, Shutdown, Refuel, Start & Hot Stby, and Run. The pistol grip locking handle assembly drives 4 banks of contacts with a total of 64 contacts.

The Mode Switch contacts enact the plant parameter logic required by the Plant Mode for Reactor Protection System (RPS) [EIS identifier: JE], Reactor Manual Control System [JD], Primary Containment Isolation System (PCIS) [JM], Refuel Interlocks [DF], Neutron Monitoring Systems [IG], and Plant Process Computer [ID] to ensure the correct setpoints and circuits are active.

Event Description

On September 26, 2022, James A. FitzPatrick Nuclear Power Plant (JAF) was reducing power for a planned plant shutdown for a refuel outage. At the time of the Scram, no structures, components, or systems were inoperable at the start of the event and contributed to the event. Control rods [AA] were being inserted and reactor power and pressure was decreasing.

At 0230, operators placed the mode switch (05A-S1) in Startup/Hot Standby (Mode 2).

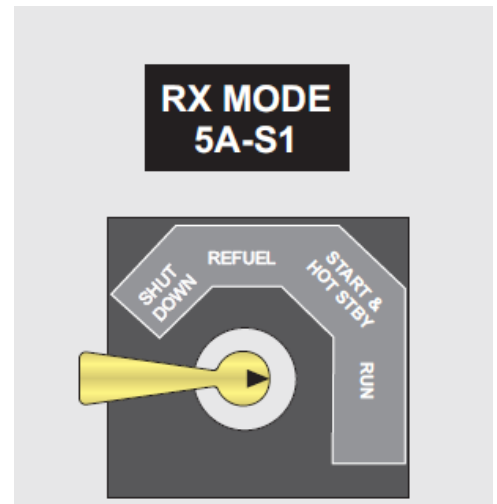
At 0306, an unplanned Scram signal was received due to a Group 1 isolation on low main steam line [SB] pressure.

All control rods were fully inserted. Main Steam Isolation Valves (MSIV) isolated except 'A' outboard MSIV (29AOV-86A) failed to close. PCIS isolated valves in multiple systems according to group 1 isolation. When the PCIS was reset, 29AOV-86A was then manually isolated.

At 0530, Operators made the decision to place Reactor Core Isolation Cooling (RCIC) system [BN] in service to augment pressure and level to assist in transition to feedwater [SJ] and condensate [SD] systems. At 0751, RCIC was shut down and in a standby lineup.

The NRC was notified of this event at 0541 by ENS 56119 in accordance with 10 CFR 50.72(b)(2)(iv)(B) RPS actuation while critical and 10 CFR 50.72(b)(3)(iv)(A) for systems actuation. A reactor engineering analysis of reactor criticality was performed for the period of time prior to the event, and it was determined that the reactor was sub-critical. On October 4, 2022, ENS 56119 was updated to retract 10 CFR 50.72(b)(2)(iv)(B).

The 'A' outboard MSIV (29AOV-86A) failed to close, but the safety function for 'A' main steam line isolation was successful because the inboard MSIV isolated successfully. Failure of the MSIV to close was documented and evaluated under IR 04524581.



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James A. FitzPatrick Nuclear Power Plant	05000 – 333	2022	– 002	– 01

Event Analysis

The Scram was initiated by an RPS actuation, which was initiated by MSIV closure, and MSIV isolation actuation was initiated by low main steam line pressure. The RPS actuation on MSIV closed position, and MSIV isolation on low main steam line pressure are not required to actuate in Mode 2 and should have been bypassed by mode switch position contacts. Mode 2 supports conditions at low reactor power, such as startup and shutdown, where the analyzed plant transient events for these required system actuations are not applicable.

The investigation confirmed the Mode Switch (05A-S1) was changed to Start & Hot Stby (Mode 2) by the Operator; however, some trip bypasses were not engaged. Moving the switch to Start & Hot Stby should have bypassed the low Main Steam Line Pressure MSIV closure and inhibit the scram on MSIV closure. However, as the plant reduced power for the planned refuel outage, the un-bypassed trip setpoints actuated Group 1 isolation valves in multiple systems, including RPS, and MSIV system actuation. The automatic actuation of these systems and manual actuation of RCIC is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

The Failure Modes Causal Team (FMCT) identified the most likely cause of failure of proper switch contacts make-up as the mode switch was operated slowly to perform detailed inspection of switch components and gearing per scheduled preventative maintenance activity. This style of operation resulted in all contacts not making up properly when the switch was operated. Contributing, the operations procedural guidance did not include validation by multiple methods, that the Mode 2 logic was properly made up following mode switch manipulation.

During Mode Switch troubleshooting the condition was not able to be repeated and no mechanical component degradation was identified. It is unlikely for multiple mechanistic failures to occur, on multiple channels, across two separate systems; therefore, the causal analysis focused on switch operation, logic validation, and corrective actions including Organizational and Programmatic changes.

Cause

The RPS and MSIV system actuation was caused by the Mode Switch 05A-S1 failing to enact the logic required to bypass the Group 1 isolation on Main Steam Line Low Pressure (PCIS) and the MSIV Closure RPS Trip when taken to the Start & Hot STBY position.

Similar Events

No similar events were identified at JAF.

FAILED COMPONENT IDENTIFICATION:

Manufacturer:	General Electric
Manufacturer Model Number:	SB-9
Manufacturer Code:	G080
Component Code:	HS
FitzPatrick Component ID:	05A-S1

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2. DOCKET NUMBER

05000 – 333

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REV
NO.

– 01

Corrective ActionsCompleted Actions

A Failure Mode Causal Team (FMCT) was formed to identify most likely causes and assist in completion of troubleshooting actions prior to plant start-up. During troubleshooting, no abnormal component degradation was identified when 05A-S1 was inspected. Prior to startup, 05A-S1 was tested satisfactorily.

Operating procedure OP-65 revision was completed prior to reactor start-up adding requirements to ensure when the mode switch position is changed, the proper method of operation and switch contact logic is validated by redundant indications.

'A' outboard MSIV (29AOV-86A) was repaired prior to startup with a new air pack and solenoid valve cluster assembly. The replaced items were sent to PowerLabs for failure analysis.

Planned Actions

Review risk analysis for worst case maintenance impacts in operational risk systems will be completed prior to next refuel outage.

Case study on event reviewed with operations during training cycle.

Safety Significance

There were no actual or potential nuclear or radiological consequences during this event.

Plant systems operated in response to this event as expected with the exception that a single component (29AOV-86A) of the MSIV system did not isolate. The redundant in inboard component fulfilled the safety function of the MSIV system. In addition, Safety/relief valves were not required to control pressure.

References

Issue Report, IR 04524574, Reactor Scram from Mode 2 During Plant Shutdown, dated September 26, 2022

Issue Report, IR 04524581, 'A' Outboard MSIV Failed to Close on GP 1 Isolation, dated September 26, 2022